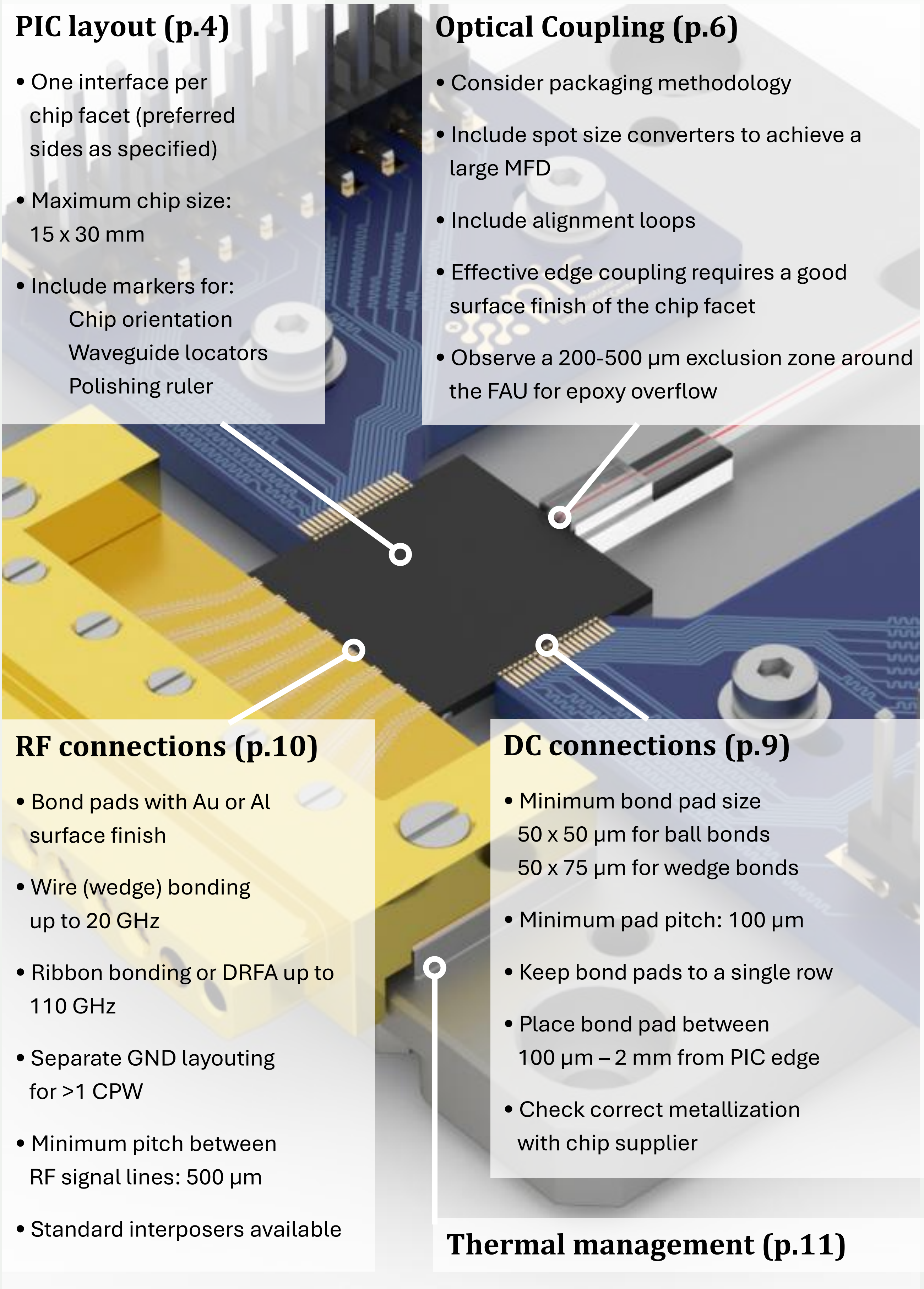
PIC design guidelines

v1.2
Swiss PIC, 17.11.2025

Contents



PIC layout (p.4)

- One interface per chip facet (preferred sides as specified)
- Maximum chip size: 15 x 30 mm
- Include markers for:
  - Chip orientation
  - Waveguide locators
  - Polishing ruler

Optical Coupling (p.6)

- Consider packaging methodology
- Include spot size converters to achieve a large MFD
- Include alignment loops
- Effective edge coupling requires a good surface finish of the chip facet
- Observe a 200-500 µm exclusion zone around the FAU for epoxy overflow

RF connections (p.10)

- Bond pads with Au or Al surface finish
- Wire (wedge) bonding up to 20 GHz
- Ribbon bonding or DRFA up to 110 GHz
- Separate GND layouting for >1 CPW
- Minimum pitch between RF signal lines: 500 µm
- Standard interposers available

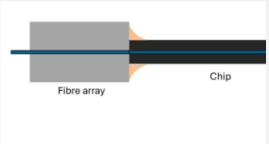
DC connections (p.9)

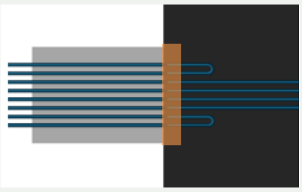
- Minimum bond pad size
  50 x 50 µm for ball bonds
  50 x 75 µm for wedge bonds
- Minimum pad pitch: 100 µm
- Keep bond pads to a single row
- Place bond pad between 100 µm – 2 mm from PIC edge
- Check correct metallization with chip supplier

Thermal management (p.11)

Quick Overview on Size Constraints

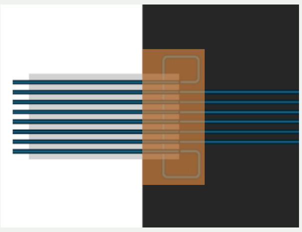
Optical interface - edge coupling





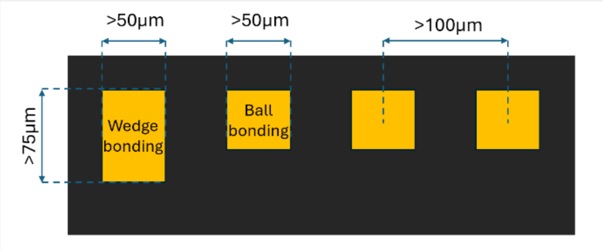
Please keep a 200 µm exclusion zone around the chip edge free from any relevant structure, as epoxy might overflow to this region.

Optical interface - grating coupling



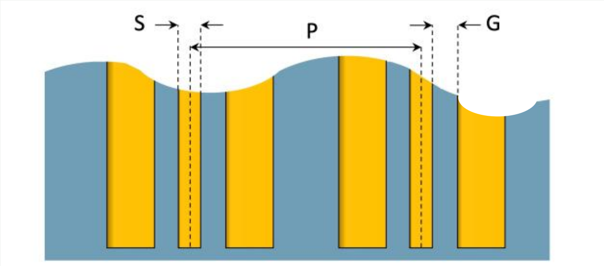
Overspill of epoxy is typically more pronounced in surface grating couplers, please respect a 500 µm exclusion zone around the size of the FAU in this case.

DC bond pads



Please respect these minimum bond pad sizes. Pads should be placed no further than 2mm from the chip edge.

RF connections



	S [um]	G [um]	P [um]
Microstrip	50	NA	500*
CPW	60	50	800*

Table 1: Minimal sizes (µm) for signal (S) width, their spacing (G) and S-to-S pitch (P).

* The values provided for channel pitch are based on assembly considerations and RF PCB manufacturing guidelines. The transmission line design is dependent on the substrate and desired cut-off frequency, hence it is preferred to be tailored individually. Please contact us to get more details.

Swiss PIC has established a few interposers with channel pitch (P) of 1 mm and developed sizes for GSG (G+S+G) summarized in Table 2. Number of RF channels: 1, 4, or 8.

Note: These offerings can cover a wide range of CPW lines; however, custom dimensions can also be developed for specific chips.

Feature	G + S + G [um]	P [um]
Fine	170	1000
Medium	300	1000
Large	430	1000

Table 2: Established CPWs for RF interposers

1 Introduction

Photonic packaging is a term that covers the optical, mechanical, thermal and electrical interfaces to a photonic integrated circuit (PIC). It remains one of the most costly and frequently underestimated steps of the PIC development process, with some estimates showing 50-70% of a PICs cost is associated with the packaging phase. If photonic packaging is not considered from the start of the PIC design phase, PICs will often not be compatible with standard packaging techniques.

At Swiss PIC we are interested in working closely with PIC designers to help them become aware of the state-of the art packaging techniques and allow them to understand and access the packaging facilities, reducing their overall development costs. As the lack of standardisation in PIC design remains an issue in the industry, this document will serve as guideline for PIC developers to ensure the transition to the packaging process is as straightforward and low-cost as possible.

There are several engineering challenges associated with the PIC packaging stage. The optical interfaces from the chip to fibre or free space generally requires components to be placed with sub-micron accuracy to avoid excessive losses of optical signal. In general, this level of precision is not achievable with passive alignment techniques. Instead, active alignment techniques are required where the optical transmission is monitored during the alignment procedure to ensure the optimal position is located. There are also frequently mismatches between the optical mode sizes of the optical fibre and the chip's waveguide which need to be addressed in the design phase to prevent high losses or expensive development solutions (Section 3). Thermal issues can also arise, particularly in high power or low temperature applications, with coefficient of thermal expansion (CTE) mismatches or temperature gradients having the potential to add undesirable stresses and shifts within the package. With the rise of ever higher frequencies within the RF domain, this is another area which can be challenging when packaging as a device's performance can be hampered by sub-optimal packaging design. Finally, the solutions developed need to be scalable for future higher-volume development phases, which should be considered when developing initial solutions.

At Swiss PIC we aim to work with PIC developers from the prototype phase. We are able to work with all major foundries, and a range of materials and wavelengths. Although the guidelines here should provide a framework to allow your chip to be compatible with standardised packaging processes, customisation is available if these cannot be met. We recommend checking the PIC design with us before manufacturing, as this can often lead to cost savings further down the line.

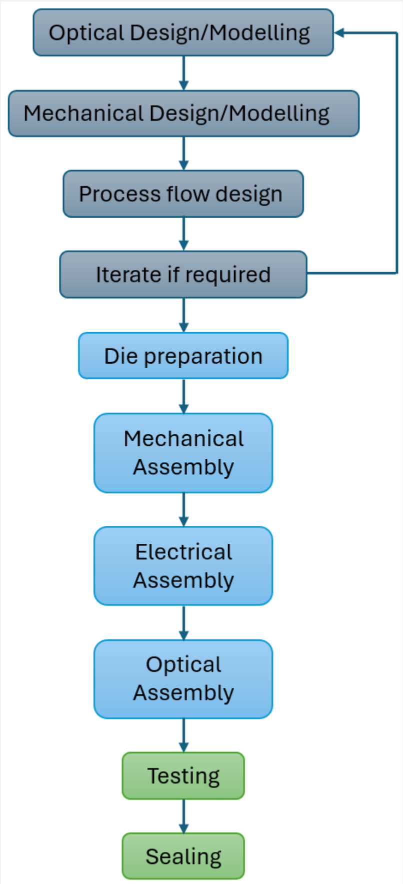


Figure 1: Graphical representation of the typical PIC packaging design flow.

2 Chip Layout

When determining the chip layout, it is preferable that each edge will contain only a single connection type. This is to prevent access issues with tooling, ensure interconnects are not impacted by subsequent processes, and allow mechanical clamping for the edge polishing of optical facets.

At Swiss PIC our standard package has a single side for optical interconnects, two sides for DC electrical, and one for RF connections. This is illustrated in Fig. 2. The maximum chip size we can accommodate is 15 x 30 mm. It should be noted that it is possible to accommodate chip designs with two optical faces, although this is non-standard.

The optical face is used primarily for fibre attaches, although it can also accommodate other functions such as free space optics if required. The electrical faces will be for wirebonds, or potentially BGA connection to a carrier. RF faces will consist of wedge/ribbon bonds for low loss impedance matched transmission.

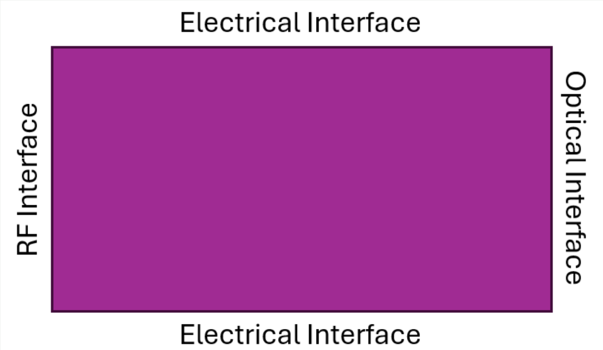
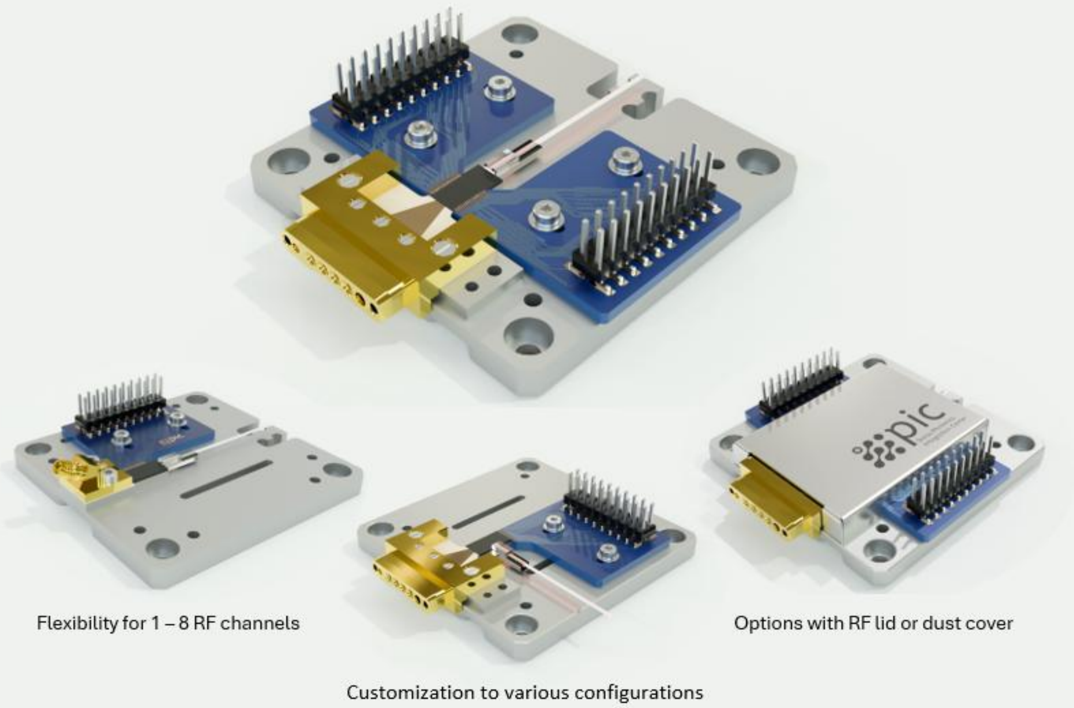


Figure 2: Chip edges should be individually assigned for different purposes. We recommend following this layout to improve compatibility with our processes.

Notes for designers:

- Each chip edge should be associated with a single interface type
- Keeping these consistent with Fig. 2 will keep development costs low
- Maximum chip size 10 x 20 mm



2.1 Alignment Markers

Many packaging processes rely on visual or computer aided alignment of components. Even for optical components where active alignment is required, an initial passive alignment is still performed to narrow down the search area, speeding up the process.

Alignment markers are required on the PIC to act as reference points during these processes.

2.1.1 Chip orientation

Markers for the chip orientation are used during the die bonding and sometimes wirebonding phase. These are usually placed in the chip corner and should be distinct and easily visible. As an example, a rectangular shape could be positioned in the chip's top left corner which could be detected by the image recognition software.

2.1.2 Waveguide locators

During the optical alignment phase waveguides are not always clearly visible from above, particularly when embedded waveguides are used. Waveguide locators are typically placed to indicate the position of the outer waveguide in an array that are used as the alignment loops (see Section 3).

2.1.3 Polishing rulers

Polishing rulers should be located by the chip edges that require mechanical polishing. These act as a guide to show the amount of material that should be removed and allow monitoring of the polishing process. A polishing ruler is shown in Fig. 4.

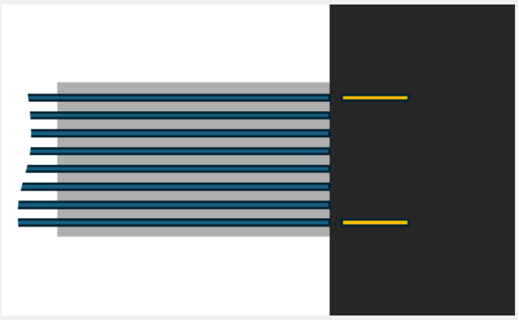


Figure 3: Alignment markers used to aid in the alignment of fibre arrays

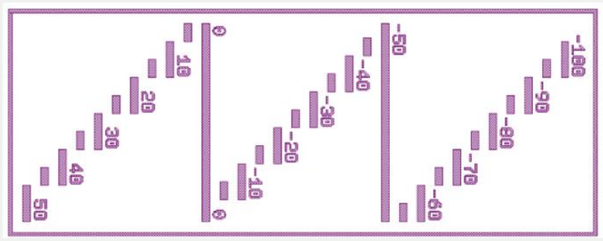


Figure 4: Polishing ruler example from PHIX

Notes for designers:

- Include chip orientation markers
- Include waveguide locators
- A polishing ruler should be placed on each edge that requires polishing

2.2 Polishing

For many packaging processes, notably edge coupling, a high-quality surface finish is desirable on the chip edge. This will greatly improve the overall coupling efficiency. The angle of the edge polish can be adjusted as desired, which can be useful on optical facets where back reflections need to be avoided. If an edge needs to be angled, we recommend doing this in the plane shown in Fig. 5. This avoids issues with waveguide and/or fibre pitch being altered along angled surfaces.

When waveguide tapers are present on the chip, a section of waveguide should be present after the taper which is at least 30 µm long, which is to be removed during polishing. If this is not present, there is a risk that the polishing process will grind away part of the taper and reduce the final mode size, in turn lowering coupling efficiency. All sides to be polished require a polishing ruler as described in section 2.1.3.

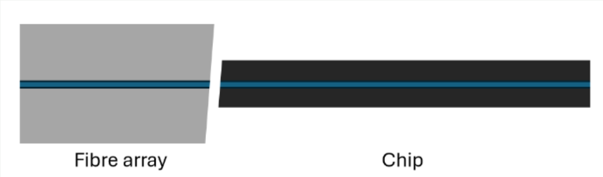


Figure 5: Chip and FVA polished at an angle to avoid back reflections, viewed from the side of the chip.

Notes for designers:

- If edge coupling is required consider dicing processes and surface finish requirements
- Facets can be polished at an angle to avoid back reflections
- A polishing ruler should be placed on each edge that requires polishing
- Leave at least 30 µm at the end of waveguide tapers if the surface requires polishing of the fibre array.

3 Optical Coupling

Optical coupling methods can be loosely grouped into two categories, active and passive alignment. Active alignment involves monitoring a feedback signal when placing the component to ensure it is located in precisely the right place, whereas passive alignment does not require this.

For prototype packaging active alignment of optical components to waveguides is almost always required at the optical interface. While passive coupling is possible, this is non-standard and can require significant application specific development work, although this can sometimes be worthwhile if very high volumes are expected.

In order to perform an active alignment procedure some way of monitoring the coupling efficiency is required, generally by maximising an optical signal on a detector. If the PIC has on-board lasers or photo diodes it can be possible to use these for this measurement, but this is undesirable as it requires the chip to be probed during alignment, and these channels may not be ideally positioned at the edges.

Instead, it is preferable that alignment loops are included in the PIC design - waveguides which connect one optical port to another for the sole use of fibre alignment.

Multiple optical fibres are typically held together using V-groove arrays (FVA), which contain 1-64 fibres dependant on application. For multi-channel arrays the standard fibre pitch is 127 or 250 µm, and will normally have a concentricity of 0.5 µm. Waveguide arrays should therefore be designed with this in mind. Where possible the optical I/O ports should be located in the middle of the chip edge in order to allow for easier tooling access. Fibres are normally attached with an optically transparent, UV curing epoxy.

It is also important to consider chip flatness as chips larger chips can bow, which will reduce the optical coupling of some channels.

The two main schemes of fibre coupling offered are edge coupling, and grating coupling.

3.1 Edge Coupling

- ✓ typically lowest losses
- ✓ broadband
- ✓ less polarisation sensitive

The fibre array will be actively aligned to the edge of the chip, UV curing epoxy applied, then cured in place.

An essential part of designing a low loss optical system is mode size matching. The mode size of the optical fibre will depend on the wavelength and specific fibre, a typical 1550 nm fibre will have a mode size of 10.5 µm. PIC waveguide sizes will depend on material and design but can be in the region of 150 x 150 nm - a significant difference. Larger mode sizes have the additional effect of being more forgiving during the alignment process. We strongly recommend designing spot size converters on edge-coupled PICs. If spot size converters are not an option on the chip, it is possible to use an interposer - a separate glass structure containing a spot size converter. However, this is more expensive and can lead to greater losses. If only a single fibre is required, then a lens on the end of the fibre can also be an option to focus the light to the correct mode size. However, these can be harder to implement on fibre arrays due to tolerance issues in the axial plane.

For edge coupling we require two alignment loops written into the PIC, one at each side of the FVA, for a total of 4 channels. This is shown in Fig. 6 (b). When attaching the fibre array via epoxy, a small amount of epoxy will seep out into the surrounding area. For this reason, there should also be an exclusion zone of 500 µm around the optical interface where other components should not be positioned.

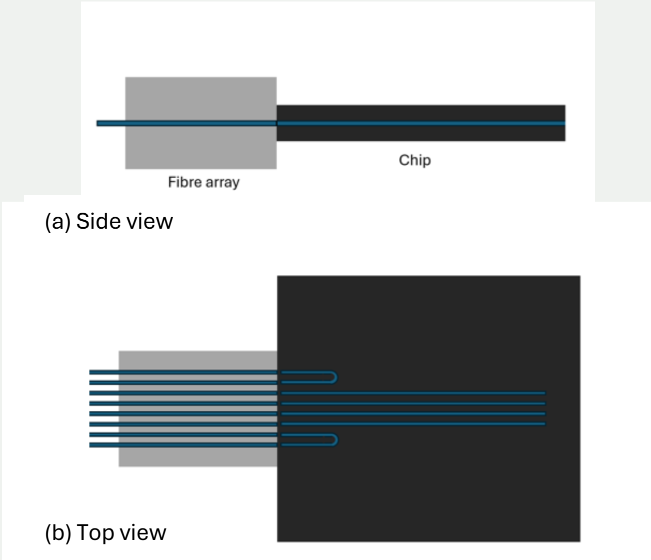


Figure 6: Simplified diagram of an edge coupling scheme (a) side view (b) top down view showing alignment loops

Notes for designers:

- All optical interfaces should be placed only along the optical edge (Fig. 2)
- Mode matching is essential for good coupling efficiency
- Spot size converters should be included in the PIC design when possible
- Effective edge coupling requires a good surface finish
- 4 additional waveguide channels should be included as alignment loops
- Leave a 500 µm exclusion zone from the facet where the epoxy will seep out

3.2 Grating Coupling

Grating coupling is an alternative scheme in which light exit/entry is not restricted to the edge of the chip. Grating coupling is typically more forgiving to misalignment than edge coupling (a 2.5 μm shift leads to around a 1dB loss), and usually has a larger contact area with the chip making it more mechanically robust. However, it tends to be less efficient than edge coupling, and is restricted by bandwidth and polarisation.

Traditional vertical coupling schemes, as shown in Fig 7 (a), are useful for probing chips but not generally used for photonic packaging due to the large vertical footprint and fragile arrangement. More typical is a quasi-planar arrangement as shown in Fig 7 (b) where the end of the fibre array is polished so that light is reflected downward into the chip, either by a high-reflective coating or through total internal reflection. For the quasi-planar approach the distance from the fibre to the grating is determined by the FVA lid height. However, it is also possible to have a lidless FVA arrangement. In this case the distance between the fibre and the grating is determined only by the thickness of the epoxy layer.

When designing grating couplers the packaging method should also be considered. Understanding the fibre's mode size, height above the chip, phase upon entry, expected insertion angle, and epoxies in the optical path will allow the grating's performance to be accurately simulated. It is also possible to design gratings which are more forgiving to misalignment (usually at the expense of peak coupling efficiency) for applications where challenging environments are expected. Typically angled FVAs are polished to an angle of around 40°, with an angular tolerance of around ~ 0.5°. Grating insertion angles that allow total internal reflection within the FVA are preferred, although reflective coatings are available on the end surface of the FVA if needed.

For grating coupling configurations a single loop back structure is generally sufficient as an alignment loop, as shown in Fig. 7 (c). This means that just 2 additional channels will be needed. The grating couplers should be at least 750 μm away from any chip edges, although a distance of 1mm is preferable. An exclusion zone of 500 μm should also be left around the gratings in order to avoid epoxy interfering with other processes.

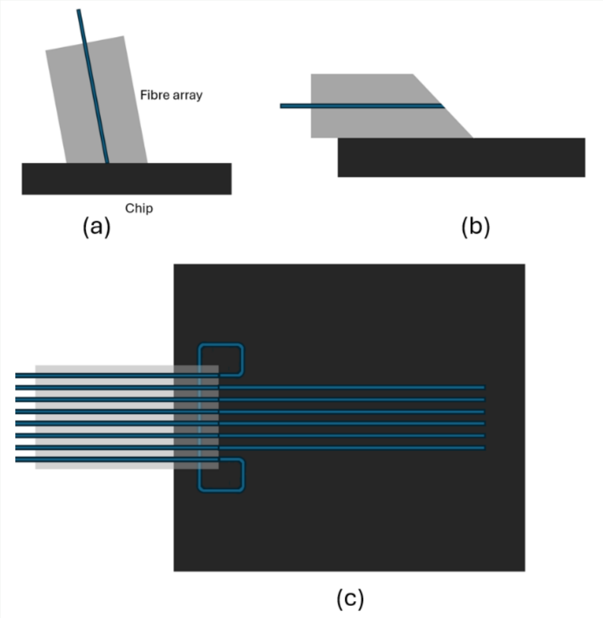


Figure 7: Grating coupler schemes (a) vertical coupling arrangement (b) quasi-planar approach (c) Top view of quasi-planar approach showing loopback structures.

Notes for designers:

- Optical interfaces should be placed only along the optical edge (Fig. 2)
- Consider packaging methodology when designing grating couplers to ensure maximal coupling efficiency
- Spot size converters should be included in the PIC design when possible
- 2 additional waveguide channels should be included as alignment loops
- Grating couplers should be at least 750 μm from the chip edge
- Leave a 500 μm exclusion zone around areas which will contain epoxy

4 Electrical Connection

4.1 DC Connections

The most common method to provide DC electrical connections to PICs is through wirebonding. When designing chips, designers should place wirebond pads on the North/South side of the PIC as shown in Fig. 2. The minimum size of the pads depends on the type of bond used. For ball bonding pads should be a minimum of 50 x 50 µm, whereas wedge bonding requires pads to be at least 50 x 75 µm due to the elongated profile of a wedge bond. In both cases there should be a minimum pitch of 100 µm as shown in Fig. 8.

Ideally the bond pads will be positioned in a single row along the edge of the chip to reduce the risk of accidental shorts between adjacent wires. In general, wirebonds should also be kept as short as possible to avoid sagging, so it is recommended not to place wirebond pads further than 2-3 mm from the chip edge. For the packaging processes there is no minimum distance between the pads and the chip edge, although bond pads that are affected by the dicing process can sometimes cause delamination. For this reason we advise leaving a gap of 100 µm between bond pads and the chip edge.

We can offer wedge, ball, and ribbon bonding as required, in both Au and Al. It is important to confirm with the PIC manufacturer that the metallization stack on these pads is suitable for wirebonding to prevent potential peeling issues.

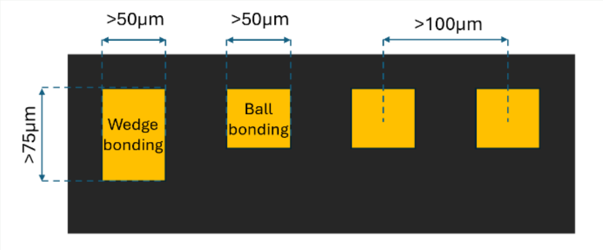


Figure 8: Minimum pitch of wirebond pads, and minimum sizes for both ball and wedge bonding

Notes for designers:

- DC wirebond pads should be placed only along the edges assigned for the electrical interface (Fig. 2)
- Minimum bond pad size is 50 x 50 µm for ball bonds and 50 x 75 µm for wedge bonds
- Minimum pad pitch of 100 µm
- Keep bond pads to a single row
- Bond pads should be no further than 2 mm from the chip edge
- Use a wire-bondable surface finish resp. metallisation stack

4.2 RF Connections

Classical wire bonding supports RF connections up to approximately 20 GHz. For higher frequencies, short bondwires with precise pad alignment or ribbon bonding is recommended. When compatible with the chip topology, flip-chip bonding provides an effective approach to minimize interconnect length and is better suited for high-volume manufacturing. Main considerations in the RF interface design are impedance matching, and minimizing dielectric and metallic losses, as well as reflections.

To ensure smooth transitions, wire or ribbon bonds between the PIC and PCB should be kept at a minimal length and as parallel as possible. This requires precise alignment of the bond pads on the PIC and the PCB. RF interposers (as depicted in Fig. 9) mediate between transmission lines on the chip and the connector sides and enable the transition of the electromagnetic mode field.

To avoid crosstalk effects, the RF traces should maintain a minimum distance. Crosstalk in transmission lines depends on the frequency, substrate thickness, dielectric constant, and signal spacing. The degree of isolation required for crosstalk is specified based on the signal type, between 20 dB to 60 dB. Evaluation of the cross talk between RF traces is possible theoretically [2–5] or empirically using calculators and simulators.

To enhance the isolation in the case of coplanar waveguides (CPWs), ground-connected vias are used on both sides of the line to promote the quasi-TEM mode. The rest of the constraints about the dimensions indicated in figure 10 are found in table 3.

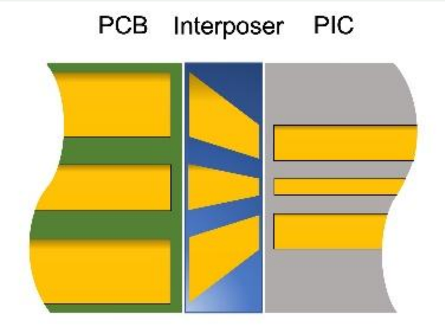


Figure 9: RF interposers to gradually match the dimensions of the transmission lines

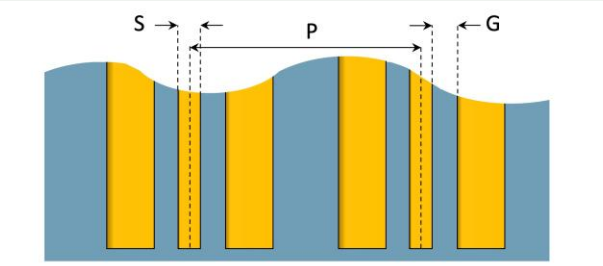


Figure 10: RF trace dimensions on PIC: signal width (S), their spacing (G) and S-to-S pitch (P)

	S [um]	G [um]	P [um]
Microstrip	50	NA	500*
CPW	60	50	800*

Table 3: Minimal values for transmission line dimensions

* The values provided for channel pitch are based on assembly considerations and average RF interposer manufacturing guidelines, while smaller dimensions are possible for custom orders. However, the pitch must be adjusted to meet the crosstalk requirements.

Notes for designers:

- Bondpads should have a wire-bondable surface finish.
- In case of using more than one CPW, grounds are recommended be layouted separately (figure 10)
- The interposer substrate can be a Ceramic or a ROGERS laminate. The track width and gap for interposers on ROGERS are recommended to be at least 100 µm.
- Consult table 3 or page 2 for the constraints of microstrips and CPWs dimensions

5 Thermal Management

As many PICs are sensitive to temperature variations or thermal gradients, thermal management is a key aspect of package design. Our standard PIC package includes a thermo-electric cooler (TEC) and thermistor for thermal control, although these can be removed or replaced with a simple heater if required. During the thermal design process we will be considering material properties such as thermal expansion and conductivity to simulate how heat and thermally included stresses will be distributed within the package, and adapt the design where problems arise.

During the thermal design we shall also be looking at bonding techniques. Typically, the options are epoxy, sintering pastes, or solders. If a strong thermal path is required, for example with higher power devices or devices which are susceptible to thermal gradients, a sintering paste or solder is often recommended. These will require the base of the chip to be metallised in a way which is compatible with the chosen bonding process to avoid issues such as scavenging.

An area of interest to Swiss PIC is packaging for cryogenic applications. These applications require special considerations for epoxy choice, bonding, and fibre attachment which will affect most aspects of the design and should be discussed early on.

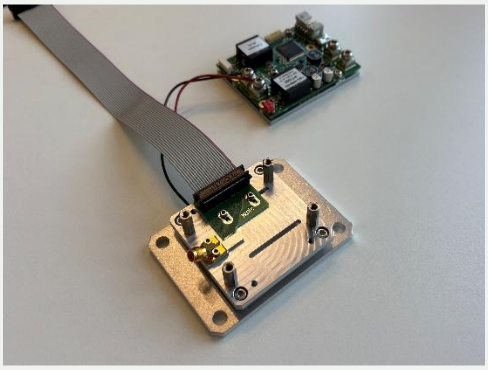


Figure 11: Prototype of the standard characterization package with TEC and TEC driver

Notes for designers:

- Thermal requirements should be highlighted early in the design process
- Metallization at the base of the chip allows more options for thermal management
- Ensure this metallization is suitable for the intended processes

References

[1] B. Wadell, “Transmission line design handbook”, Artech House, 1991.

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[4] “IPC 2221A Generic Standard on Printed Board Design” Northbrook, IL: Association Connecting Electronics Industries (IPC), 1998.

[5] “IPC D-317A-1995 Design Guidelines for Electronic Packaging Utilizing High-Speed Techniques” Northbrook, IL: Association Connecting Electronics Industries (IPC), 1995.